

Poster: What Can Review with Security Concern Tell Us before Installing Apps?

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ABSTRACT

Providing pertinent information to potential users about security concerns drawn from user feedback is essential to enhance the security level of the entire mobile ecosystem. According to a usable security survey, star ratings and reviews can help users' download and permission decisions. However, many apps have security problems with a high star rating. To confirm this problem, we build a dataset by collecting 56,439,878 star ratings and reviews for 8,999 popular game apps on Google Play Store. This study proposes a model using an active learning method that distinguishes Review with Security Concern (RSC) from user reviews. As a result, we find the star rating and RSC have a very weak correlation (Pearson correlation value -0.2949). Thus, it shows that the current star rating system does not reflect users' security concerns and needs improvements to deliver a more practical metric before installing an app by users.

CCS CONCEPTS

• **Security and privacy** → *Human and societal aspects of security and privacy*; • **Computing methodologies** → *Artificial intelligence*.

KEYWORDS

user feedback, star rating, security and privacy concern, active learning, Convolutional Neural Network, Natural Language Processing

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1 INTRODUCTION

Nowadays, as mobile apps are widespread in every domain, the threat from mobile is also increasing. Security concerns are essential when users decide not to download an app [1]. Even popular apps like TikTok and Facebook continue to face security issues. Unfortunately, it is almost impossible for ordinary users who are not professionals in security to notice a cyber threat of a mobile app without external assistance until it is installed and the app executes a malicious action. Therefore, people look up the app's reputation toward from other users' appraisals when they decide to install the app and grant permissions to it [5]. Most App stores, such as Google Play, have review systems where users can share their experience with apps and rate them. The reviews can contain positive or negative opinions, including security concerns [4]. However, for most people, it is a hassle task that checks security concerns from others' reviews whenever they download an app. According to Shen *et al.*'s survey [7], star ratings and reviews can help users' download and permission decisions. If an app with security issues is getting high stars, that is a problem. Investigating whether a star rating could indicate security concerns has great implications for designing indicators that provide future security feedback.

In a broad sense, we defined all evil actions that could cause time or money damage to users as security concerns. (1) DATA LEAKAGE, (2) EXCESSIVE ADVERTISING and (3) INEXPEDIENT PAYMENT. DATA LEAKAGE means leaking user data or accessing data that does not require. EXCESSIVE ADVERTISING stands for too many ads that users can no longer use the app. INCONVENIENT PAYMENT refers to a case where a user is forced to pay after installation or the payment provided is different from the information provided before payment. Here are some examples of app reviews:

"... people can see the data / information without asking permission to the phone number owner. Scammer can use the data for fraud purpose ..."

"My kid loves this game but they keep on showing ads even after I have paid for no ads. I don't know if this is deliberate or a glitch."

2 EXPERIMENT

Data Collection. In the mobile app market, game apps have higher revenues [3] and higher user interactions than other apps [6]. Therefore, game apps are suitable for our experiments as they are apps with high threats and active feedback. We developed a

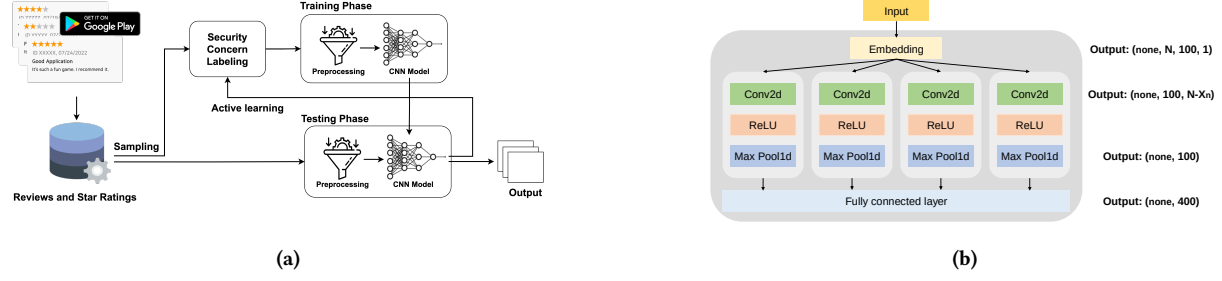


Figure 1: (a) The overall framework structure of our proposed method using active learning, (b) Structure of CNN model that classifies whether the review includes users' security concerns

Table 1: Results of our model on labeled data

	Precision	Recall	F1-score
Not-RSC	0.9725	0.9615	0.9669
RSC	0.7344	0.7966	0.7642

crawler and collected reviews written after 2020 for Android game apps with over 10,000,000 downloads. We collected a total of 8,999 apps and 56,439,878 reviews and star ratings. We manually labeled the 6,502 reviews for a month to train the classifier. The dataset we built is available at: [2]

Model. Figure 1(a) shows the overall process. For the first, our model classifies reviews after training with the label. Second, our model does preprocessing with tokenization and stopword removal. Third, vectorization was performed using global vectors for word representation (Glove). For the CNN model, we set the embedding dimension to 100. Thus, a sentence containing N -words is embedded as an $N \times 100$ vector. The structure of the model used for classification was used by increasing the window size to 3 for layer1, 4 for layer2, 5 for layer3, and 6 for layer4 using one convolutional layer in parallel, as shown in Figure 1(b). Each vector has a size of 100 through a 1D max pooling layer. The concatenated vector of size 400 put the fully connected layer.

Active Learning. For higher performance, additional labeling was performed on data that were not correctly classified when classified. The newly labeled data has been added back to the training set.

Results. Our model identifies the Review with Security Concern (RSC) among the reviews. We divided the labeled reviews into train, valid, and test dataset at a ratio of 8:1:1. Table 1 shows the performance of our model. The proposed model successfully detects 4,128,951 RSCs out of 56,439,878 reviews.

3 ANALYSIS OF THE RESULT

Figure 2 shows the correlation between the proportion of RSC in all reviews and star ratings. The Pearson correlation value is -0.2949; and the p-value is less than 0.0001. It indicates the result is statistically significant. The results show that the correlation between the two variables is very weak. More specifically, apps with low RSC have little regularity, and apps with high RSC are too few.



Figure 2: Correlation between the proportion of Reviews with Security Concern in all reviews and star rating

4 CONCLUSION

To summarize, the key contributions of this paper are as follows: 1) We analyze whether the current star rating is ineffective for users to consider security issues before downloading an app. 2) We build a large dataset from the real world and propose a model to identify RSC using the active learning method. As a result, our proposed model shows a high performance and identified RSC from the dataset. Unfortunately, the correlation between the proportion of RSC and the star rating is very weak. Thus, we can conclude the current star rating system does not help let users know about possible security concerns before installation. During the decision-making process to install an app, if a user can access a more practical metric such as the proportion of RSC, it will be helpful to make a better decision.

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